

14-2a A Simple Example of Profit Maximization

Let’s begin our analysis of the firm’s supply decision with the example in [Table 2](#). Column (1) in the table shows the number of gallons of milk the Vaca Family Dairy Farm produces. Column (2) shows the farm’s total revenue, which is \$6 times the number of gallons. Column (3) shows the farm’s total cost. Total cost includes fixed costs, which are \$3 in this example, and variable costs, which depend on the quantity produced.

Table 2						
Profit Maximization: A Numerical Example						
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Quantity	Total Revenue	Total Cost	Profit	Marginal Revenue	Marginal Cost	Change in Profit
(Q)	(TR)	(TC)	(TR – TC)	(MR = ΔTR/ΔQ)	(MC = ΔTC/ΔQ)	(MR – MC)
0 gallons	\$ 0	\$ 3	–\$3			
				\$6	\$2	\$4
1	6	5	1			
				6	3	3
2	12	8	4			
				6	4	2
3	18	12	6			
				6	5	1

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Quantity	Total Revenue	Total Cost	Profit	Marginal Revenue	Marginal Cost	Change in Profit
(Q)	(TR)	(TC)	(TR – TC)			(MR – MC)
4	24	17	7			
				6	6	0
5	30	23	7			
				6	7	–1
6	36	30	6			
				6	8	–2
7	42	38	4			
				6	9	–3
8	48	47	1			

Column (4) shows the farm's profit, which is computed by subtracting total cost from total revenue. If the farm produces nothing, it has a loss of **\$3** (its fixed cost). If it produces 1 gallon, it has a profit of **\$1**. If it produces 2 gallon, it has a profit of **\$4** and so on. Because the Vaca family's goal is to maximize profit, it chooses to produce the quantity of milk that makes profit as large as possible. In this example, the Vaca Farm maximizes profit by producing either 4 or 5 gallon of milk for a profit of **\$7**.

There is another way to look at Vaca Farm's decision: The Vacas can find the profit-maximizing quantity by comparing the marginal revenue and marginal cost of each unit produced. Columns (5) and (6) in [Table 2](#) compute marginal revenue and marginal cost from the changes in total revenue and total cost, and column (7) shows the change in profit for each additional gallon produced. The first gallon of milk the farm produces has a marginal revenue of **\$6** and a marginal cost of **\$2**; hence, producing that gallon increases profit by **\$4**

(from $-\$3$ to $\$1$). The second gallon produced has a marginal revenue of $\$6$ and a marginal cost of $\$3$, so that gallon increases profit by $\$3$ (from $\$1$ to $\$4$). As long as marginal revenue exceeds marginal cost, increasing the quantity produced raises profit. Once the Vaca Farm has reached 5 gallons of milk, however, the situation changes. The sixth gallon would have a marginal revenue of $\$6$ and a marginal cost of $\$7$, so producing it would reduce profit by $\$1$ (from $\$7$ to $\$6$). As a result, the Vacas do not produce beyond 5 gallons.

One of the *Ten Principles of Economics* in [Chapter 1](#) is that rational people think at the margin. We now see how the Vaca Family Dairy Farm can apply this principle. If marginal revenue is greater than marginal cost—as it is at 1, 2, and 3 gallons—the Vacas should increase the production of milk because it will put more money in their pockets (marginal revenue) than it takes out (marginal cost). If marginal revenue is less than marginal cost—as it is at 6, 7, and 8 gallons—the Vacas should decrease production. By thinking at the margin and making incremental adjustments to the level of production, the Vacas end up producing the profit-maximizing quantity.

Chapter 14: Firms in Competitive Markets: 14-2a A Simple Example of Profit Maximization

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